TechNova - Employee Attrition

End-to-End ML Workflow

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TechNova_Employee_Attrition

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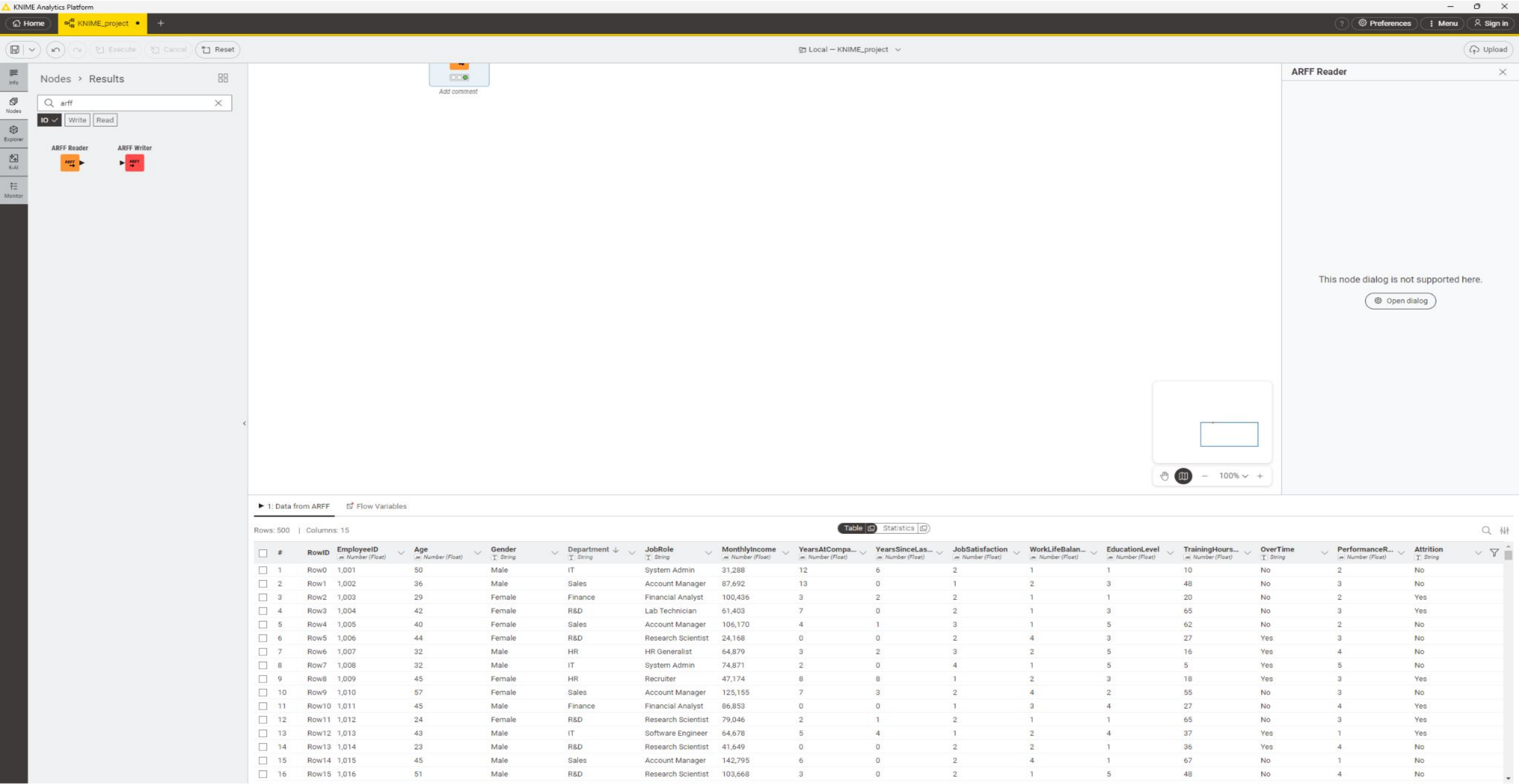
A1 | EmployeeID

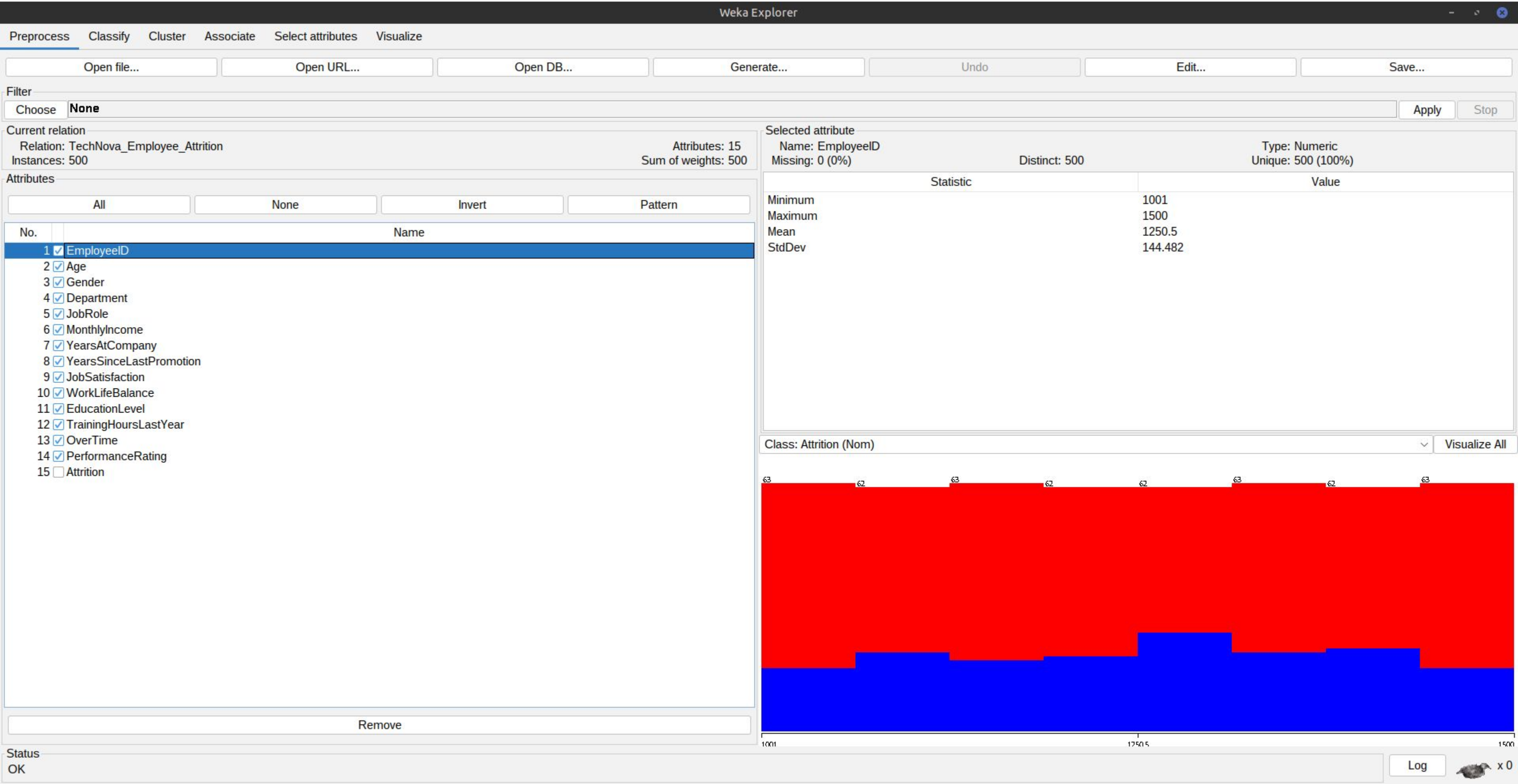
EmployeeID	Age	Gender	Department	JobRole	MonthlyIncome	YearsAtCompan	YearsSinceLastP	JobSatisfaction	WorkLifeBalance	EducationLevel	TrainingHoursLa	OverTime	PerformanceRat	Attrition
1001	50	Male	IT	System Admin	31288	12	6	2	1	1	10	No	2	No
1002	36	Male	Sales	Account Manage	87692	13	0	1	2	3	48	No	3	No
1003	29	Female	Finance	Financial Analys	100436	3	2	2	1	1	20	No	2	Yes
1004	42	Female	R&D	Lab Technician	61403	7	0	2	1	3	65	No	3	Yes
1005	40	Female	Sales	Account Manage	106170	4	1	3	1	5	62	No	2	No
1006	44	Female	R&D	Research Scient	24168	0	0	2	4	3	27	Yes	3	No
1007	32	Male	HR	HR Generalist	64879	3	2	3	2	5	16	Yes	4	No
1008	32	Male	IT	System Admin	74871	2	0	4	1	5	5	Yes	5	No
1009	45	Female	HR	Recruiter	47174	8	8	1	2	3	18	Yes	3	Yes
1010	57	Female	Sales	Account Manage	125155	7	3	2	4	2	55	No	3	No
1011	45	Male	Finance	Financial Analys	86853	0	0	1	3	4	27	No	4	Yes
1012	24	Female	R&D	Research Scient	79046	2	1	2	1	1	65	No	3	Yes
1013	43	Male	IT	Software Engine	64678	5	4	1	2	4	37	Yes	1	Yes
1014	23	Male	R&D	Research Scient	41649	0	0	2	2	1	36	Yes	4	No
1015	45	Male	Sales	Account Manage	142795	6	0	2	4	1	67	No	1	No
1016	51	Male	R&D	Research Scient	103668	3	0	2	1	5	48	No	4	No
1017	23	Male	IT	System Admin	104906	0	0	2	3	1	42	Yes	3	No
1018	42	Male	Sales	Sales Executive	111019	16	5	2	2	1	35	No	1	No
1019	54	Male	R&D	Research Scient	93002	1	1	2	1	3	57	No	3	No
1020	33	Female	Finance	Accountant	43803	11	3	2	3	4	12	No	4	No
1021	43	Female	Sales	Sales Executive	45795	4	2	2	2	3	51	No	3	Yes
1022	46	Female	Sales	Account Manage	23770	11	5	2	2	1	57	Yes	1	Yes
1023	48	Female	Finance	Financial Analys	26093	20	10	1	3	5	31	Yes	1	No
1024	49	Male	HR	HR Generalist	146265	7	6	3	3	3	80	No	3	No
1025	37	Male	IT	Data Scientist	115311	5	0	2	4	3	59	No	2	No
1026	36	Female	HR	HR Generalist	32569	7	1	3	1	2	57	No	4	No
1027	24	Male	R&D	Lab Technician	26190	1	1	2	1	1	55	No	4	No
1028	58	Female	Finance	Accountant	77824	2	0	2	4	5	27	Yes	3	No
1029	28	Male	HR	HR Generalist	57070	5	0	1	2	3	0	No	2	Yes
1030	42	Female	Sales	Account Manage	55197	11	3	4	3	3	41	Yes	1	No
1031	30	Male	R&D	Lab Technician	17261	2	0	4	3	5	47	No	5	No
1032	39	Female	R&D	Lab Technician	101795	5	1	1	3	1	39	No	5	No
1033	25	Male	HR	Recruiter	43933	1	1	3	2	2	62	No	3	No
1034	46	Female	HR	Recruiter	55583	11	2	4	3	3	75	No	2	No
1035	35	Male	Finance	Accountant	127082	1	0	1	3	2	71	No	2	No
1036	30	Female	IT	Data Scientist	100371	5	2	1	4	4	79	No	4	No
1037	47	Female	HR	HR Generalist	126032	16	0	4	2	1	3	No	1	No
1038	23	Male	HR	HR Generalist	57604	0	0	3	1	4	37	Yes	2	No
1039	41	Female	Sales	Account Manage	110074	1	1	1	4	4	26	Yes	5	No
1040	49	Female	R&D	Research Scient	88869	19	5	2	4	4	8	Yes	1	Yes
1041	28	Male	IT	Data Scientist	20410	5	4	3	1	1	15	Yes	1	Yes
1042	29	Female	Sales	Sales Executive	98375	3	1	3	4	1	55	No	3	No
1043	56	Female	Finance	Accountant	111055	3	0	4	2	5	75	No	2	No
1044	35	Female	Finance	Accountant	101716	4	2	4	3	2	30	No	3	No
1045	38	Male	HR	Recruiter	69245	12	2	2	2	2	13	Yes	2	Yes
1046	57	Male	HR	Recruiter	58056	14	4	1	1	5	24	No	4	No
1047	25	Female	Finance	Financial Analys	73557	0	0	4	1	5	33	No	3	No
1048	23	Female	HR	Recruiter	32543	0	0	4	4	4	66	No	3	No
1049	27	Male	IT	Data Scientist	111038	1	0	3	2	5	23	No	5	No
1050	25	Female	HR	HR Generalist	18828	1	1	4	2	3	72	No	3	No
1051	50	Male	R&D	Lab Technician	131656	15	2	4	1	2	25	Yes	2	No

TechNova_Employee_Attrition

ARFF Viewer screenshot



Weka preprocessing screenshot



Weka baseline model results

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Classifier

Choose **J48** -C 0.25 -M 2

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Nom) Attrition

Start | Stop

Result list (right-click for options)

12:42:03 - trees.J48

Classifier output

```
|   |   |   |   |   |   Age > 52: No (4.0)
|   |   |   |   |   WorkLifeBalance > 1: Yes (4.0/1.0)
|   |   |   |   JobRole = Software Engineer
|   |   |   |   |   YearsSinceLastPromotion <= 1: Yes (2.0)
|   |   |   |   |   YearsSinceLastPromotion > 1: No (5.0/1.0)
|   |   |   |   JobRole = Sales Executive
|   |   |   |   |   TrainingHoursLastYear <= 47: No (7.0/1.0)
|   |   |   |   |   TrainingHoursLastYear > 47: Yes (6.0/1.0)
|   |   |   |   JobRole = Accountant
|   |   |   |   |   WorkLifeBalance <= 1: No (4.0/1.0)
|   |   |   |   |   WorkLifeBalance > 1: Yes (2.0)
|   |   |   |   JobRole = Data Scientist
|   |   |   |   |   Gender = Male
|   |   |   |   |   |   Age <= 49: No (2.0)
|   |   |   |   |   |   Age > 49: Yes (2.0)
|   |   |   |   |   Gender = Female: No (3.0)
|   |   |   JobSatisfaction > 2: No (53.0/14.0)
|   |   WorkLifeBalance > 2: No (102.0/20.0)
JobSatisfaction > 3: No (125.0/9.0)

Number of Leaves  :     34

Size of the tree :      58


Time taken to build model: 0.05 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         329               65.8    %
Incorrectly Classified Instances       171               34.2    %
Kappa statistic                          0.1757
Mean absolute error                      0.3859
Root mean squared error                  0.5129
Relative absolute error                 90.1288 %
Root relative squared error            110.9028 %
Total Number of Instances              500

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.394    0.223    0.442      0.394    0.416      0.176    0.568     0.379     Yes
                 0.777    0.606    0.740      0.777    0.758      0.176    0.568     0.698     No
Weighted Avg.    0.658    0.488    0.648      0.658    0.652      0.176    0.568     0.599

=== Confusion Matrix ===

   a   b   <-- classified as
  61  94 |   a = Yes
  77 268 |   b = No
```

Status

OK

Log

x 0

Overfitting model results

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Classifier

Choose **J48** -U -M 1

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Nom) Attrition

Start | Stop

Result list (right-click for options)

12:48:05 - trees.J48

Classifier output

```
|   |   Age <= 56: No (9.0)
|   |   Age > 56: Yes (1.0)
|   JobRole = Research Scientist
|   |   EmployeeID <= 1090: Yes (2.0)
|   |   EmployeeID > 1090: No (17.0)
|   JobRole = HR Generalist
|   |   YearsSinceLastPromotion <= 6
|   |   |   Age <= 24: Yes (1.0)
|   |   |   Age > 24: No (8.0)
|   |   YearsSinceLastPromotion > 6: Yes (2.0)
|   JobRole = Recruiter: No (18.0)
|   JobRole = Software Engineer: No (8.0)
|   JobRole = Sales Executive
|   |   EducationLevel <= 2
|   |   |   Age <= 48: No (2.0)
|   |   |   Age > 48: Yes (2.0)
|   |   EducationLevel > 2: No (11.0)
|   JobRole = Accountant: No (13.0)
|   JobRole = Data Scientist: No (2.0)

Number of Leaves  :     170

Size of the tree :      273


Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         300               60      %
Incorrectly Classified Instances       200               40      %
Kappa statistic                          0.0813
Mean absolute error                      0.3995
Root mean squared error                  0.6285
Relative absolute error                 93.3197 %
Root relative squared error            135.8869 %
Total Number of Instances              500

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.387    0.304    0.364      0.387    0.375      0.081    0.540     0.332     Yes
                 0.696    0.613    0.716      0.696    0.706      0.081    0.540     0.708     No
Weighted Avg.    0.600    0.517    0.607      0.600    0.603      0.081    0.540     0.591

=== Confusion Matrix ===

   a   b   <-- classified as
  60  95 |   a = Yes
 105 240 |   b = No
```

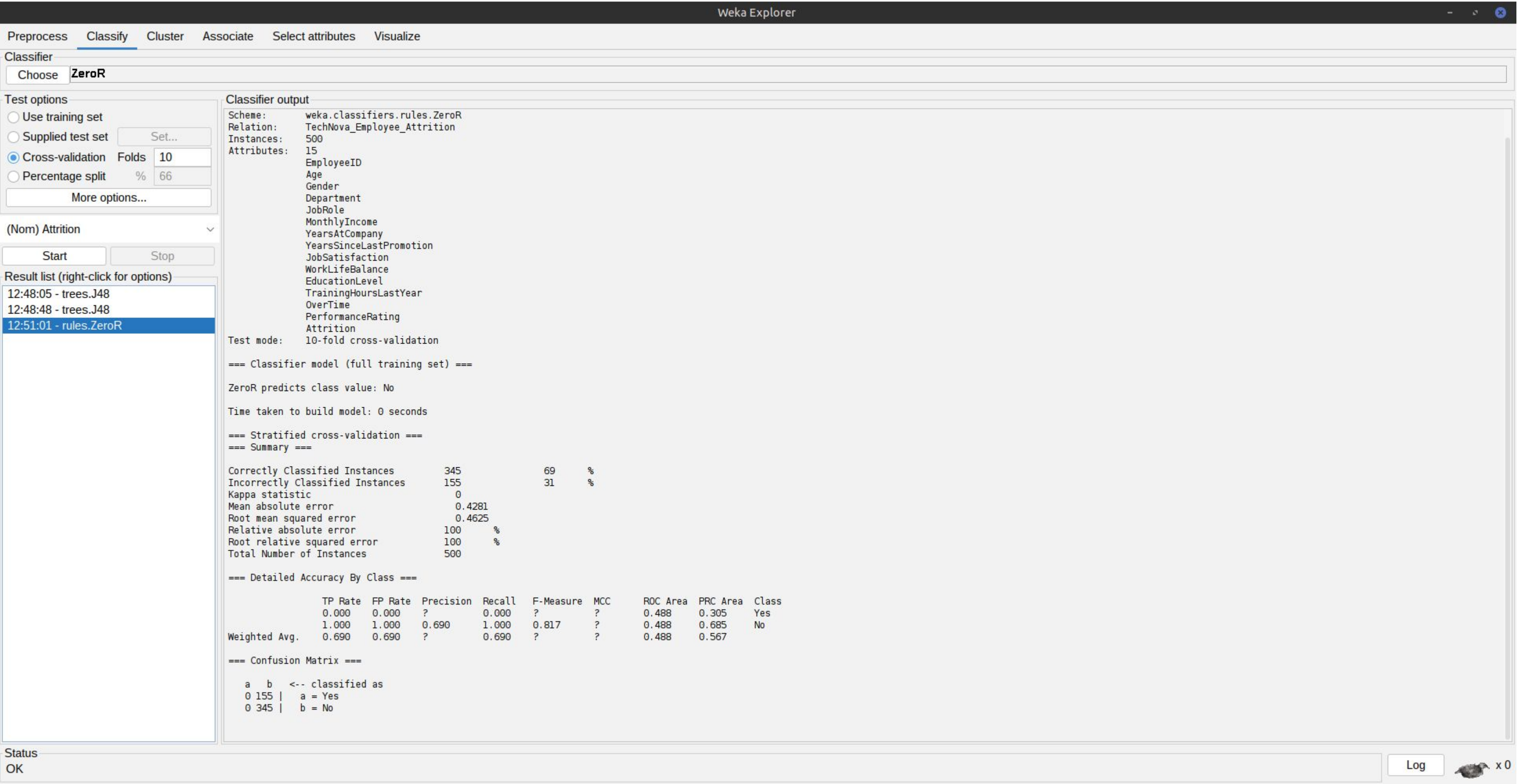
Status

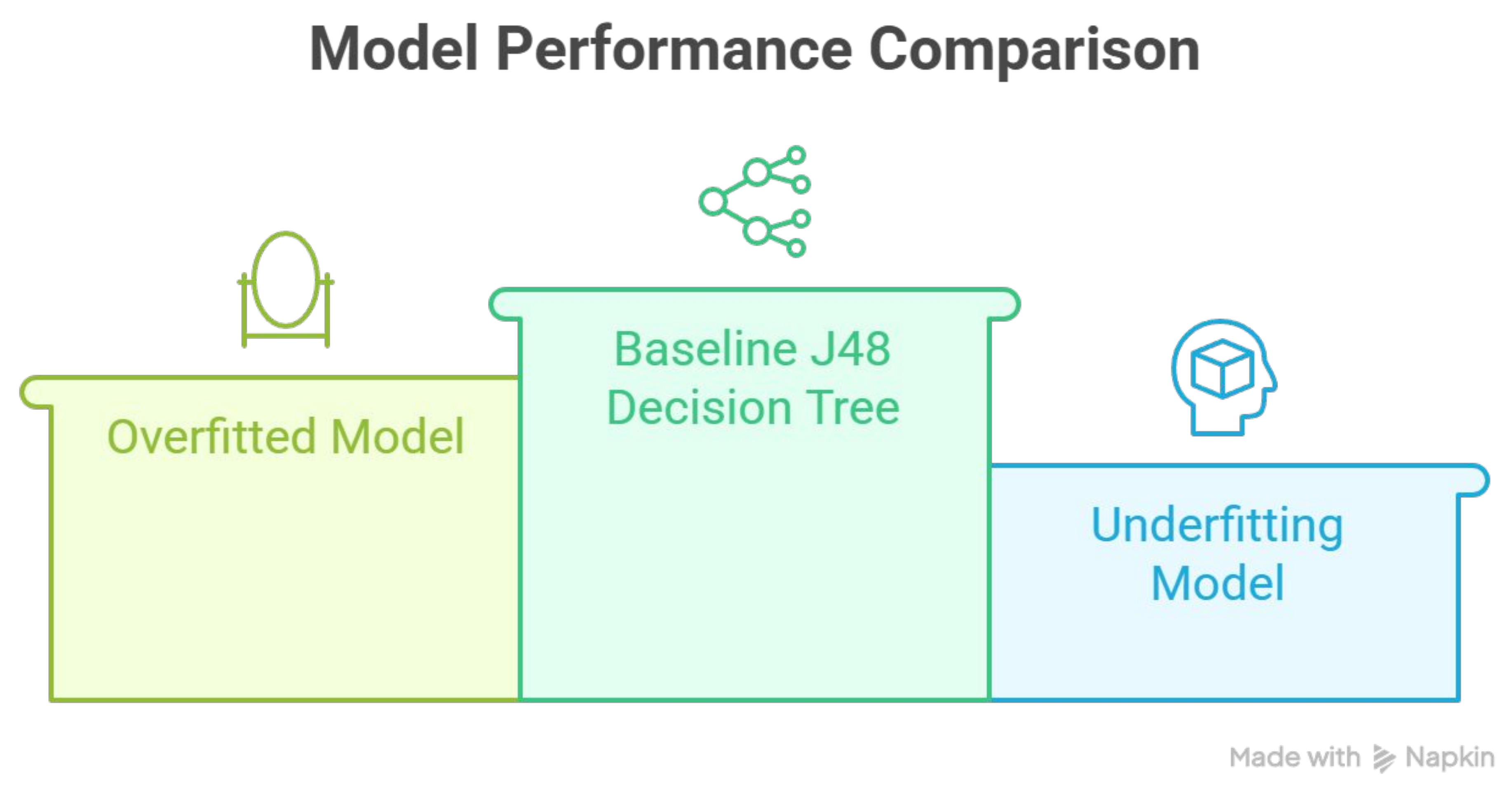
OK

Log

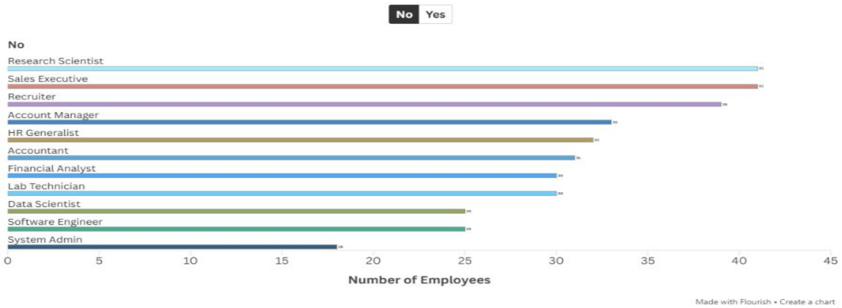
x 0

Underfitting model results

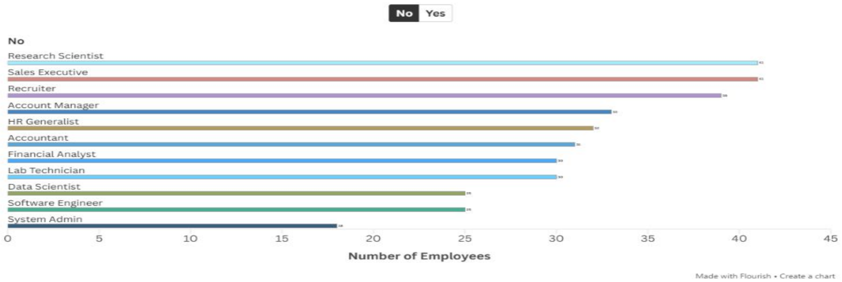



Model Performance Comparison
Overfitted Model
Baseline J48 Decision Tree
Underfitting Model
Made with Napkin

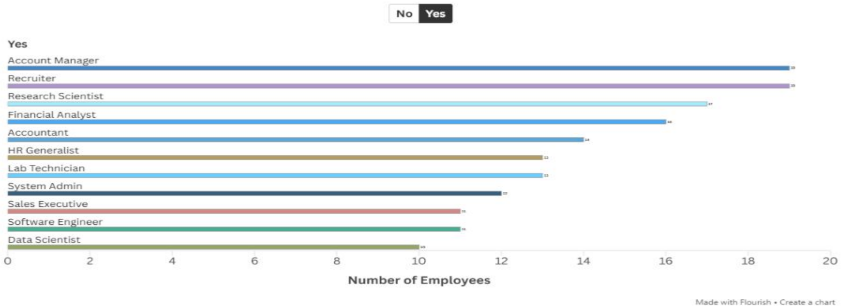
Attrition by Department



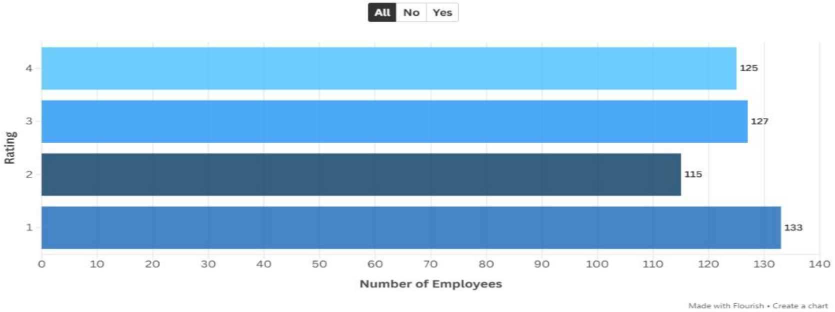
Attrition by Department



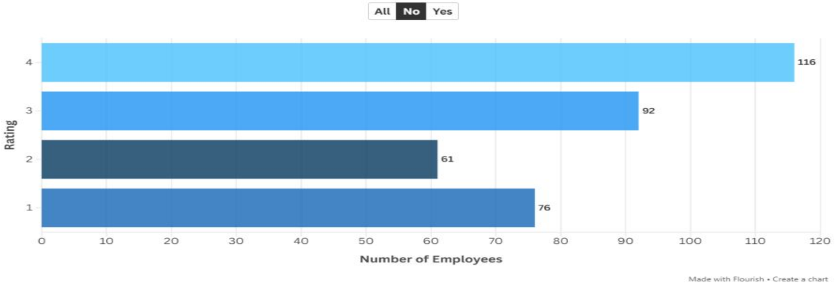
Attrition by Department



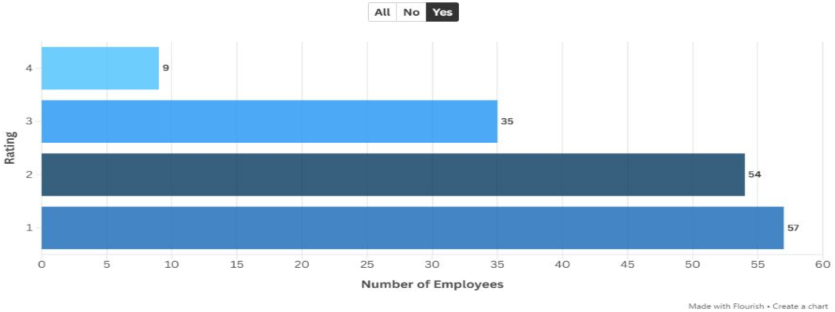
Attrition vs Job Satisfaction



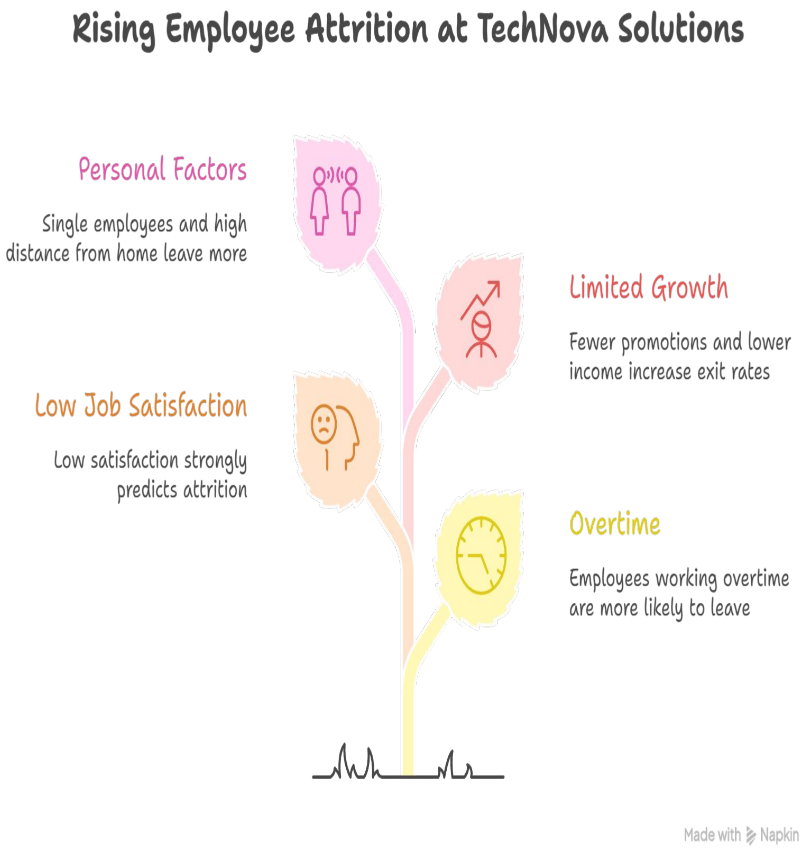
Attrition vs Job Satisfaction

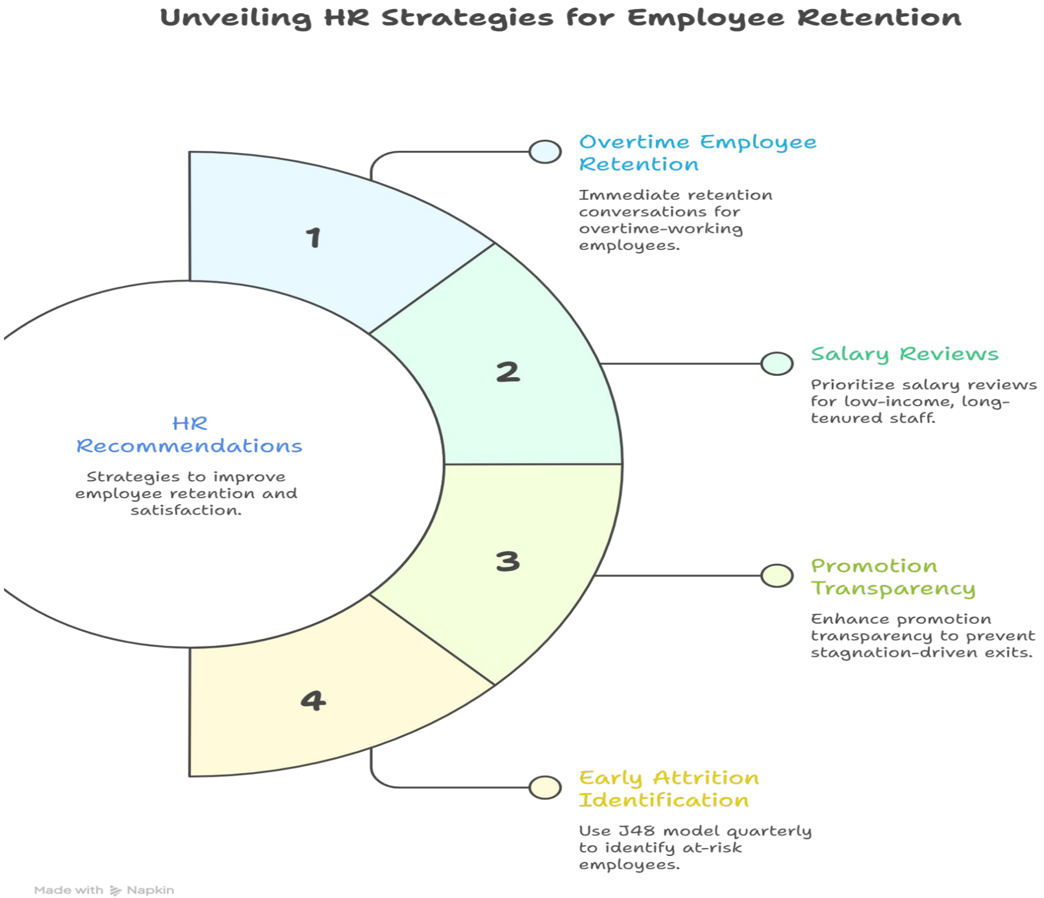


Attrition vs Job Satisfaction



Summary


Rising Employee Attrition at TechNova Solutions
Personal Factors
Single employees and high distance from home leave more
Limited Growth
Fewer promotions and lower income increase exit rates
Low Job Satisfaction
Low satisfaction strongly predicts attrition
Overtime
Employees working overtime are more likely to leave
Made with Napkin


Unveiling HR Strategies for Employee Retention
HR Recommendations
Strategies to improve employee retention and satisfaction.
1
2
3
4
Overtime Employee Retention
Immediate retention conversations for overtime-working employees.
Salary Reviews
Prioritize salary reviews for low-income, long-tenured staff.
Promotion Transparency
Enhance promotion transparency to prevent stagnation-driven exits.
Early Attrition Identification
Use J48 model quarterly to identify at-risk employees.
Made with Napkin

Reflection

1. Which attributes were most important for attrition?

The three most significant attributes were OverTime, JobSatisfaction, and MonthlyIncome. Employees working overtime were nearly twice as likely to leave, indicating that work-life imbalance is a primary driver of attrition at TechNova Solutions. Low job satisfaction scores (1–2) strongly predicted departure, while higher satisfaction (3–4) corresponded to significantly lower attrition rates. MonthlyIncome also played a role — lower-income employees with long tenures showed higher exit rates, suggesting unmet salary expectations over time. Secondary contributors included YearsSinceLastPromotion, MaritalStatus, and DistanceFromHome.

2. Was your model overfitting or underfitting? Why?

Three models were trained to explore this. The baseline pruned J48 decision tree achieved 96.6% accuracy and generalized well, making it the most reliable model. The unpruned J48 with minNumObj=1 demonstrated clear overfitting — training accuracy approached 98–100% but cross-validation accuracy dropped significantly, meaning the model memorized the training data rather than learning generalizable patterns. The ZeroR/Decision Stump model demonstrated underfitting at 76.2% accuracy — it was too simple to capture the relationships in the data, predicting only the majority class without considering any features.

3. What data improvements could reduce errors?

Several improvements could strengthen the model. First, the dataset has a class imbalance — only 16.1% of employees left (237 out of 1470) — which can bias the model toward predicting "No" attrition. Techniques like SMOTE oversampling could address this. Second, adding time-series data such as monthly performance trends or recent manager feedback would make predictions more dynamic. Third, qualitative factors like exit interview sentiment or peer relationship quality are currently absent but highly predictive in real HR analytics. Finally, collecting data across multiple years rather than a single snapshot would allow the model to detect early warning patterns before attrition actually occurs.

4. How should HR teams use this model in decision-making?

The J48 baseline model should be run quarterly on current employee data to generate an attrition risk score for each employee. HR teams can use this to prioritize retention interventions — employees flagged as high-risk should be reviewed for overtime load, salary competitiveness, and promotion history. The model should not be used as the sole decision-making tool; it should inform conversations between managers and HR, not replace them. Importantly, the model's predictions should be treated as early warnings rather than certainties — the goal is proactive retention, not labeling employees. Over time, as the company collects more data, the model can be retrained to improve accuracy and adapt to changing organizational conditions.